

College Football Totals: Is There a Real Edge, and What Should Beat Vegas Become?

Executive conclusion

Yes, there is credible evidence that exploitable edges can exist in college football totals. But the evidence does not support the idea that “college football totals” as a whole are broadly soft or that simply moving from first-half Unders to full-game Overs and Unders will create an advantage. The strongest academic evidence points in almost the opposite direction: the overall college-football totals market is reasonably efficient, closing totals are more accurate than opening totals, and naïve strategies such as blindly betting Unders, following reverse line movement, or fading public Over money are unlikely to constitute a durable system. ¹

That is not bad news for Beat Vegas. It actually tells us what the system would need to become.

The opportunity is not **“predict whether the game goes Over or Under better than everyone else.”** The opportunity is to identify **specific situations in which the market systematically misprices scoring**, and then only bet when the estimated error is large enough to overcome vig and uncertainty. Published college-football research has documented conditional inefficiencies involving weather, travel, rule changes, and a particularly interesting structural bias in team totals. Those findings demonstrate that an efficient market in the aggregate can still contain profitable conditional pockets. ²

My conclusion for Beat Vegas is therefore:

Do not abandon the first-half Under system and replace it with a generic full-game totals model. Instead, evolve Beat Vegas into a college-football scoring and totals engine with multiple market-specific strategies competing against one another.

The existing first-half Under strategy should remain the **incumbent/champion** until a challenger proves superior. The first challenger I would research is **two-sided first-half totals**, because it reuses most of the information and logic you already have while removing the artificial assumption that only Unders can contain value. The second challenger should be **full-game Over/Under**, built on the same underlying scoring engine but treated as a separate market with separate probabilities, thresholds, and validation.

The long-run version of Beat Vegas should not say, “We are an Under system.” It should say:

We estimate how many possessions and points a college-football matchup should produce, compare that distribution with the sportsbook market, and wager only when the market is materially wrong.

That distinction is enormous.

My confidence levels after reviewing the evidence are roughly:

Question	Research conclusion
Is the overall full-game CFB totals market easy to beat?	No
Is there evidence of conditional inefficiencies?	Yes
Is there evidence that blindly betting Unders works?	No
Is there evidence that blindly following line movement works?	No
Does weather appear potentially underpriced?	Yes — strong evidence
Does travel appear potentially underpriced?	Yes — moderate evidence
Can rules/regime changes temporarily create mispricing?	Yes — strong historical evidence
Are opening numbers potentially more exploitable than closing numbers?	Yes, but not automatically
Are closing totals generally more accurate?	Yes
Is first-half CFB demonstrably softer than full-game?	Not established by published research
Should Beat Vegas abandon first-half Unders now?	No
Should Beat Vegas research both sides and full games?	Absolutely
Could a sophisticated totals system have a sustainable edge?	Plausibly yes, but it must be demonstrated prospectively

The key is that **the edge, if we find it, is likely to be conditional, selective and time-sensitive—not a universal Over/Under bias.**

What the research actually says about college-football totals

The most useful broad study I found is James Francisco and Evan Moore's work on the “wisdom of crowds” in college-football betting. They analyzed Division I games from the 2003-04 through 2015-16 seasons using Sports Insights opening and closing lines. Their totals sample contained 7,557 games with usable opening and closing totals. The result is important for Beat Vegas: at the opening line, Unders covered 50.09%; at the closing line, Unders covered 50.14%. Statistically, they could not reject an efficient-market result. In other words, there was no broad “just bet the Under” advantage hiding in the data. ³

That same research also found something even more important: **the market gets smarter as kickoff approaches.** The closing total was closer to the actual game score than the opener in 4,032 observations, or 53.35% of cases where the comparison could be made. Mean absolute deviation fell from 13.28 points for opening totals to 13.08 for closing totals, and the difference was statistically significant. The authors

concluded that opening and closing totals were generally efficient, but the closing number was a better forecast. ¹

That gives us a foundational principle for Beat Vegas:

The sportsbook total should not be treated as an enemy prediction to ignore. It should be treated as one of the strongest predictive features available.

Instead of trying to forecast a game from zero and declaring:

Model = 51.2
Sportsbook = 56.5
Bet Under

the more sophisticated question is:

What information would justify believing that the market's 56.5 is 3-5 points wrong, despite the considerable information already embedded in 56.5?

That is a much harder problem—but also a much more realistic one.

The opening number is not the same thing as the closing number

The fact that the closing market improves on the opener matters tremendously. Circa's own description of line-setting explains the mechanism: an opening sportsbook puts up a number, typically at lower limits, professional bettors attack numbers they believe are wrong, and the bookmaker reacts. Other sportsbooks then enter around the emerging market consensus. As additional books and bettors participate, the consensus becomes more established. Circa says it publishes its first weekly college-football lines on Sunday and explicitly describes opening limits as generally lower because early prices help the sportsbook obtain information from professional bettors. ⁴

So there probably **is** more opportunity early.

But “early lines are softer” does not mean:

Bet everything Sunday.

It means:

If Beat Vegas possesses information or a model the market has not fully incorporated, Sunday through the early week is when that information is most valuable.

There is a trade-off. Early in the week you have potentially worse market prices but less complete information regarding weather, availability and injuries. Late in the week you know more, but the market does too. ⁵

That suggests Beat Vegas ultimately needs two forecasts:

An Originator forecast designed to attack early numbers, and a **Closer forecast** that incorporates late weather, personnel information and market movement.

They should not necessarily produce the same bets.

Why market efficiency does not kill the project

A 50/50 aggregate market can still contain conditional edges.

Imagine:

- Weather games: Unders 56%.
- Certain matchup types: Overs 55%.
- Everything else: almost exactly 50%.

Aggregate everything together and the inefficiencies disappear statistically.

That is precisely why the conditional findings in the literature are interesting. The broad market may be efficient while particular subsets are not. The recent weather study, travel research, rule-change evidence and team-total work all point toward that possibility. ²

This distinction should become central to Beat Vegas:

We are not looking for proof that totals are inefficient. We are looking for repeatable conditions under which the market's error distribution shifts enough to become bettable.

There is no durable generic Under bias in the strongest long sample

This matters because your present system is first-half Unders.

The 2003–2015 full-game sample does **not** show a general full-game Under phenomenon. Its approximately 50.1% Under result is effectively neutral before vig. ³

At standard -110 pricing, break-even is approximately **52.38%**.

For perspective:

- 52.38% $\approx$ break-even.
- 54% generates about a 3.09% expected return on amount risked at constant -110.
- 55% generates about a 5.0% expected return.
- 56% generates about a 6.91% expected return.

So moving from roughly 50% to a legitimate 54–55% out-of-sample system would be an enormous achievement.

That is also why a backtest showing 57–60% should immediately trigger skepticism rather than celebration. The first questions should be leakage, line timing, stale prices, selection bias, multiple testing and whether the quoted line could actually have been bet.

Where a genuine totals edge is most likely to live

The most interesting part of the research is that there **are** documented examples in which college-football betting markets appear not to incorporate information perfectly.

I would divide the possible Beat Vegas advantages into four evidence tiers.

Weather is one of the strongest candidates

A 2024 paper by Steven Salaga and William Howley specifically examined weather in the NCAA football totals market. They found statistically significant relationships between weather characteristics and totals outcomes, while those weather variables did **not** significantly explain the magnitude of totals line movements. Their interpretation was that the market systematically undervalued weather information. They also reported that simple strategies based on only three weather characteristics could produce profitability in their sample. ⁶

That is unusually relevant to what you are trying to build.

The important lesson is not “wind = Under.” A simplistic weather filter would likely be arbitrated eventually.

The deeper opportunity is interactions:

Weather × offensive style × passing rate × explosiveness × kicking × stadium × total × market movement.

For example, identical weather does not necessarily have identical scoring implications for a run-heavy offense and an offense built around deep passing. That is a modeling inference rather than a published result, but it is exactly the kind of conditional relationship worth testing given the paper's evidence that weather has historically not been perfectly assimilated into totals prices. ⁶

Weather should therefore be a **core Beat Vegas information layer**, not a small bonus variable.

Travel deserves considerably more attention than most handicapping systems give it

B. Jay Coleman's peer-reviewed study examined whether college-football spreads and totals accurately account for visiting-team travel, including distance, direction, time zones, temperature changes, elevation and aridity. The study concluded that the market was an inaccurate and inefficient processor of some travel effects, with the strongest recurring results appearing in particular late-season underdog travel situations. ⁷

That does not mean Beat Vegas should implement a rule such as:

West-to-East = Under.

It suggests something much better: build a proper **travel stress score** and test whether the sportsbook's implied scoring expectation systematically misses at its extremes.

Possible inputs worth investigating include:

- distance traveled;
- time zones crossed;
- eastward versus westward movement;
- local kickoff time relative to a team's home body clock;
- short week versus normal rest;
- consecutive road games;
- elevation differential;
- seasonal timing.

Those should be tested as continuous and interacting variables rather than converted immediately into betting "systems." Coleman's findings justify the research question; they do not justify a universal travel bet.

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Rules changes are a fascinating source of temporary edge

There is historical precedent for NCAA rule changes breaking the scoring market temporarily.

Research on the 2006 NCAA clock changes found that scoring declined and that the market did not immediately adjust enough, producing significant returns to Under betting. Rule changes in 2007 then produced additional market volatility that diminished as the season progressed. 9

This is extremely relevant because college football underwent another meaningful timing change in 2023. For Divisions I and II, the clock began continuing after most first downs, stopping for a first down only during the final two minutes of each half. The NCAA itself said the change was expected to modestly reduce the number of plays. 10

This creates a warning for any Beat Vegas backtest:

A 2017 college-football game and a 2025 college-football game should not automatically be treated as observations from the same scoring environment.

And it matters even more for **first-half totals**, because the exact number of available possessions in the first 30 minutes is directly influenced by timing rules.

Models should therefore explicitly distinguish scoring regimes rather than simply dumping ten years of games into one training set.

This could also create a repeatable class of edge:

Market adaptation to structural changes.

Rule changes, technology, substitutions, overtime procedures, offensive trends and other structural shocks can move the true scoring process before bookmakers have enough current-season data to recalibrate perfectly. The 2006 evidence shows that this has happened before. ¹¹

Team totals contain one of the most interesting documented inefficiencies

This was not one of your three original options, but it may become strategically important.

Robert Arscott's 2023 *Journal of Sports Economics* paper examined college-football team totals together with point spreads. His argument is structural: spread and team-total markets jointly imply scoring expectations, but team scores cannot fall below zero. That censoring can create bias in market-implied scoring distributions. A very simple strategy built only from the betting lines reportedly exceeded a **55% win rate over roughly two decades**, enough in the study to overcome typical wagering costs. ¹²

This does **not** prove that Beat Vegas should become a team-total system.

It proves something more intellectually important:

There can be mathematical inconsistencies between related college-football markets even when the headline market appears efficient.

That opens an entirely different Beat Vegas research direction later:

- full-game total,
- point spread,
- home team total,
- away team total,
- first-half spread,
- first-half total,

should be viewed as a **connected system of prices**, not isolated bets.

If one market implies Team A should score 34.2 and Team B 17.8 while another set of prices implies materially different distributions, Beat Vegas could eventually detect the inconsistency.

That may ultimately be more powerful than trying to build a raw game-score predictor from scratch.

Public Over bias is real behavior, but not necessarily free money

Research using actual NCAA football wagering percentages found that bettors were more likely to choose the Over as posted totals became higher and were more likely to bet Overs in televised games. However, the same research found evidence that sportsbooks can deliberately price around obvious bettor preferences rather than merely trying to balance equal money on both sides. ¹³

This is critical.

The statement:

“The public loves Overs.”

can be true.

While this statement:

“Therefore Unders are profitable.”

is false.

Sportsbooks are aware of bettor behavior. In situations where preferences are predictable, the bookmaker may price the bias into the line. ¹³

So betting splits should be used as **information**, not as an automatic contrarian trigger.

Reverse line movement is not enough

Reverse line movement is popular among betting communities because it seems to reveal where “sharp money” is positioned.

The empirical evidence is much less exciting. Francisco and Moore report that reverse line movement appeared in roughly 20–25% of their college-football totals observations, but their related research found that following it was **not generally a profitable totals strategy**. ¹⁴

That means Beat Vegas should track RLM, but not worship it.

The interesting features are more granular:

- Which book moved first?
- At what limit?
- Was the move 0.5 or 4 points?
- Did the entire market follow?
- Did price change before number changed?
- Was weather released?
- Was a quarterback ruled out?
- Did the total move but the spread remain stable?
- Did first-half and full-game totals move proportionally?
- Did the move reverse later?

Line movement should become model input, not a betting system.

Conference trends and rivalry trends are much weaker candidates

There is literature finding inefficiencies in certain college-football interconference spread situations, which establishes that conference perception can affect market pricing. But that evidence concerns point spreads and does not demonstrate a robust totals effect. ¹⁵

I would therefore be very cautious about rules such as:

Big Ten game = Under
Rivalry = Under
SEC = Under
MAAction = Over

Those patterns are exactly the type most vulnerable to data mining.

Conference should absolutely exist in the model because it proxies schedule strength, style and opponent ecosystems. Rivalry status can also be encoded as a contextual variable. But neither deserves a standalone betting rule until a clean out-of-sample test establishes one.

The same caution applies to conference versus nonconference games.

FBS versus FCS and smaller schools could be valuable—but this remains a hypothesis

It is plausible that low-profile games are less efficiently priced because information is thinner, limits can be lower and fewer professional bettors specialize in them. Sportsbook discussion of smaller and niche markets generally supports the idea that information can be harder to acquire in less prominent markets. But I did **not** find high-quality CFB-specific evidence proving that FCS/FBS totals reliably yield superior betting returns. ⁴

That distinction is important.

I would rank FCS mismatches as a **high-interest research hypothesis, not a known edge**.

There are reasons they might be attractive:

less information, enormous talent gaps, unusual rotation patterns, greater uncertainty about starters, and potentially less sophisticated market attention.

There are also reasons they may be terrible:

poor data quality, limited historical lines, low limits, unpredictable garbage-time behavior and difficulty estimating FCS opponent strength.

For Beat Vegas, those games should probably get **more uncertainty**, not automatically more confidence.

First-half Unders versus full-game totals

This is where the research becomes less definitive.

I found a substantial academic literature on full-game college-football betting markets, spreads and team totals. I did **not** find a comparable body of peer-reviewed research establishing whether college-football **first-half totals** are more or less efficient than the full-game total.

First-half CFB totals are widely offered by current sportsbooks, but their existence is much easier to document than their historical efficiency. ¹⁶

That means anyone confidently saying:

“First-half totals are softer.”

or:

“Full-game totals are sharper.”

is going beyond what the published CFB evidence can demonstrate.

We need to test it ourselves.

Why your first-half Under system could genuinely have an advantage

There are logical reasons a first-half system could work.

The first half removes much of the late-game state dependence that affects full games. A full-game total can be transformed by score differential, desperation fourth downs, prevent defense, backups, end-game clock management and overtime. First-half wagers finish before most of those second-half states occur. This does not mathematically guarantee less variance, but it means the process being modeled is different.

Your current system could also benefit from specialization. There is substantial value in knowing one market deeply instead of being mediocre across every available market.

And first-half scoring is highly sensitive to something we can attempt to estimate:

How many meaningful possessions will occur before halftime?

That naturally leads toward a possession model.

Why limiting the system to Unders may eventually become a weakness

There is no compelling evidence that “Under” itself is a permanent college-football anomaly.

Suppose Beat Vegas calculates:

Expected first-half points: 31.4

Market: 27.5

If the model is properly calibrated and trustworthy, refusing to bet the Over because Beat Vegas is philosophically an “Under system” would be irrational.

A genuine pricing model should not care which side it bets.

The key word is **eventually**.

You should not immediately make the model symmetrical simply because symmetry sounds sophisticated. It is entirely possible that the features that produce a good Under model are not equally effective on Overs.

Unders can be driven by:

- possession suppression,
- adverse weather,
- offensive mismatch,
- slow pace,
- poor finishing,
- offensive injuries.

Overs can be driven by an entirely different distribution:

- explosive-play vulnerability,
- defensive busts,
- tempo interaction,
- short fields,
- aggressive fourth-down behavior,
- special-teams scoring,
- mismatch-driven efficiency.

So I would treat **1H Over** as a new model class even if it shares the same base scoring engine.

Full-game totals give us one enormous advantage: data

The best broad research dataset discussed above contained 7,557 opening/closing totals across thirteen seasons. Full-game prices are relatively straightforward to archive, compare and validate. ¹⁷

The modern open-source CFB ecosystem is also good. `cfbfastR` exposes college-football play-by-play and links to games, drives, statistics, metrics, ratings, betting, recruiting, coaching and venue information; it also maintains expected-points and win-probability tools. CollegeFootballData's current API provides game, team, player, recruiting, ranking and analytics data. ¹⁸

That makes full-game model research dramatically easier.

Historical **first-half market prices**, especially timestamped open/midweek/close prices across multiple books, are much harder to obtain consistently.

That may be the single biggest practical obstacle to scientifically proving whether your existing 1H system is superior.

Full game also introduces information your existing model may not need

The full-game total is effectively a mixture of two different processes:

First-half scoring environment

plus

second-half scoring conditional on the first-half game state.

If Georgia leads 35-3 at halftime, the second half is not the same probabilistic game that existed at 0-0.

Therefore, an excellent full-game model should probably not simply estimate:

Team A points + Team B points.

It should model game-state pathways.

For example:

Competitive game path → normal fourth-quarter aggression

versus

blowout path → backups / clock bleed / altered play selection

versus

trailing favorite path → tempo and passing increase

versus

one-score ending → timeouts / fourth downs / clock stops / overtime possibility

This is one reason I would not automatically expect your existing 1H Under logic to transfer cleanly to full games.

The most interesting idea may be combining them

I think this is ultimately the highest-upside direction.

Instead of thinking:

1H total OR full-game total

think:

1H total + full-game total + spread + team totals = one market-implied scoring model.

Imagine:

Full game: 58.5

First half: 30.5

That means the market is assigning approximately 52% of its expected regulation scoring to the first half, although direct proportional comparisons need adjustment for second-half game state and market conventions.

Now suppose Beat Vegas believes:

First half fair: 26.7

Full game fair: 57.9

That is interesting.

It says the model does **not** hate total scoring nearly as much as the first-half price suggests.

The bet could therefore be:

1H Under, not full-game Under.

Conversely:

1H fair: 29.9 vs market 30

Full fair: 64 vs market 58.5

might indicate:

No first-half edge, but full-game Over.

This relative-market information could become one of the most distinctive parts of Beat Vegas.

Rather than simply assigning games letter grades, it could answer:

Where in the game does the market appear wrong?

That is a much stronger concept.

What a serious Beat Vegas totals model would need to understand

I would build the research philosophy around a simple decomposition:

Expected scoring $\approx$ expected possessions $\times$ scoring efficiency per possession

Everything else feeds one of those two components or affects uncertainty around them.

The cfbfastR ecosystem already provides play-by-play, drive and expected-points infrastructure that makes this approach feasible. ¹⁹

Possessions and tempo

Raw points per game are too blunt.

Beat Vegas should ultimately understand:

- plays per drive,
- seconds per snap,
- offensive pace in neutral situations,
- pace when leading,
- pace when trailing,
- run/pass tendencies,
- fourth-down aggressiveness,
- drive-start field position,
- opponent pace interaction,
- how quickly possessions actually terminate.

Two teams averaging 30 points per game can create completely different totals profiles.

One might generate 13 possessions at mediocre efficiency.

Another might generate nine possessions at exceptional efficiency.

The matchup implications are different.

Offensive and defensive efficiency

The engine should separate team quality from opponent quality.

Useful families of information include:

- expected points added;
- success rate;
- explosiveness;
- early-down efficiency;
- passing versus rushing efficiency;
- pressure and sack behavior;
- finishing drives;
- third/fourth-down tendencies;
- field position;
- kicking;
- opponent-adjusted versions of the above.

cfbfastR specifically maintains expected-points and EPA tools alongside play-by-play data, providing the type of infrastructure required to derive these features consistently. ¹⁹

The guiding rule should be:

Do not ask how many points a team scores. Ask why it scores those points.

Personnel and injuries

This is one area where public historical datasets become more difficult.

Quarterback availability obviously deserves attention in a scoring model, but Beat Vegas should go beyond a binary starter-in/starter-out flag. The appropriate question is:

What is the estimated change in offensive scoring efficiency given this personnel change?

The same applies to offensive-line injuries, multiple missing receivers, defensive-back injuries and depth problems.

I did not find strong peer-reviewed CFB totals evidence establishing a simple profitable injury rule, so injuries belong in the model as **information whose price impact must be learned**, not as another handicapping superstition.

An even more interesting research question is:

Does the sportsbook overreact or underreact to highly visible quarterback news relative to lower-profile cluster injuries?

That is precisely the kind of hypothesis Beat Vegas could test.

Coaching should be represented as behavior, not reputation

Rather than saying:

Coach X is an Over coach,

measure:

- neutral pace;
- pass rate;
- fourth-down aggressiveness;
- tempo when trailing;
- tempo when leading;
- scoring attempts near halftime;
- tendency to remove starters;
- play-calling inside opponent territory.

That converts coaching into repeatable data.

Coordinator changes also provide another possible regime break: historical team statistics may cease to describe the current offense after a major scheme change.

Strength of schedule and opponent adjustment are mandatory

College football's enormous talent dispersion makes unadjusted statistics particularly dangerous.

Putting up 7.8 yards per play against weak opponents is not equivalent to doing it against elite defenses.

Recruiting/talent data, ratings and opponent-adjusted metrics are available in modern CFB data ecosystems such as CollegeFootballData and cfbfastR. ¹⁸

For Beat Vegas, opponent adjustment should happen before any grade is considered trustworthy.

Mismatches need their own nonlinear treatment

A 42-point favorite is not simply a slightly stronger version of a 7-point favorite.

Huge mismatches introduce unique full-game issues:

- starters may leave;
- offensive aggression can fall;
- backups introduce variance;
- the underdog's expected scoring can approach the lower boundary of the distribution;
- the favorite may be able to determine late-game pace.

Arscott's censoring research is particularly interesting here because it shows that the zero boundary can create systematic problems in implied team scoring expectations. ¹²

That argues for explicitly modeling each team's scoring distribution rather than assuming both teams' points behave symmetrically around an average.

Market information itself should be a feature

This is one of my strongest recommendations.

Input variables should eventually include:

Opening

- total;
- spread;
- moneyline;
- team totals;
- 1H total.

Movement

- current total;
- total points moved;
- speed of move;
- price movement before number movement;
- cross-book disagreement;
- sharp-book versus recreational-book difference.

Closing

- sharp consensus total;
- closing spread;
- no-vig pricing.

Because closing totals have historically been more accurate than openings, Beat Vegas should learn from the market rather than pretend the market does not exist. ²⁰

I would actually favor modeling:

Actual total – market-implied total

rather than starting with:

Actual total.

That turns Beat Vegas into a **market-error model**.

That is exactly what we care about.

Line shopping and price matter

A model that predicts 55% Under probability is not automatically profitable at any price.

At -110, 55% is attractive.

At -125, the break-even probability is 55.56%.

The same prediction would have negative expected value.

This is also why closing-line-value calculations have to be handled carefully. Professional-betting platform Unabated points out that beating a -110 closing line does not automatically mean the bet was positive EV. To estimate edge from closing price properly, the closing comparison should be made against a **vig-free** benchmark. ²¹

Beat Vegas should therefore eventually grade:

Under 55.5 -105

differently from:

Under 55.5 -120.

The number alone is not the bet.

Number + price = bet.

How I would prove whether Beat Vegas actually has an edge

This is where most betting systems fail.

The purpose of the research phase should not be to produce an impressive historical win percentage. It should be to try as hard as possible to **destroy the proposed edge**.

If the edge survives, then it becomes interesting.

Start with the sportsbook as the benchmark

Every Beat Vegas model should have to defeat increasingly difficult benchmarks:

Benchmark	Question
Historical scoring average	Are we better than something trivial?
Team-strength model	Are football statistics adding signal?
Opening sportsbook total	Are we better than the bookmaker's first estimate?
Current market total	Are we better than the information currently available?
Closing sharp-market total	Did our earlier prediction contain information before the market did?

Given the historical evidence that closing totals outperform opening totals, beating an arbitrary baseline is not enough. ³

The real standard is:

Can Beat Vegas repeatedly identify errors in a highly informed market before those errors disappear?

Historical testing must be chronological

Do not random-split college-football games into training and test sets.

That lets future football regimes leak backward.

I would use walk-forward testing:

Train through Season A → test Season B.

Then:

Train through Seasons A+B → test Season C.

And continue.

That simulates the information environment in which bets would actually be placed.

Given the documented effect of football rule changes and the NCAA's 2023 timing change, era sensitivity should be explicitly tested. ²²

Every variable needs an information timestamp

This is absolutely essential.

Suppose the game kicks Saturday at 7:30 p.m.

If we are pretending the bet was placed Tuesday at noon, the model may use only information known by Tuesday at noon.

Not:

- Saturday injury confirmation;
- final weather observations;
- Saturday closing line;
- final depth chart released afterward;
- retroactively corrected statistics.

Otherwise the backtest contains lookahead bias.

Beat Vegas should effectively have historical “as-of” states:

Sunday model

Tuesday model

Thursday model

Saturday morning model

pregame model

That would let us answer one of your most important questions scientifically:

When is the best time to bet?

Do not measure success with win percentage alone

Each strategy should be evaluated across several dimensions.

Betting performance

- win rate;
- push rate;
- ROI;
- average price paid;
- maximum drawdown.

Prediction performance

- mean absolute error of projected total;
- probability calibration;
- Brier/log-loss for Over/Under probabilities.

Market performance

- average points of closing-line value;
- vig-free closing-price value;
- percentage of bets beating close;
- movement immediately following entry.

A system winning 54% while repeatedly beating the closing market is much more believable than one winning 58% while consistently getting a worse number than the close.

The latter may simply have been lucky.

But CLV is not perfect either. The closing market itself can occasionally be wrong, and in smaller markets the “close” may be less informative. That is why CLV, actual ROI and probability calibration must be evaluated together. Unabated's discussion also emphasizes that CLV must be compared appropriately against vig-free market prices rather than interpreted mechanically. ²¹

Every supposed edge needs an ablation test

Suppose the completed model goes 56%.

Remove weather.

Maybe it becomes 54.8%.

Remove travel.

54.6%.

Remove market information.

51.9%.

Now we understand where the actual predictive value originates.

That is far more useful than having an opaque 56% model.

Every subgroup needs to survive independently

Suppose Beat Vegas discovers:

Wind + slow offenses + total above 55 = Under 61%.

Do not immediately trust it.

Ask:

- Does it work in 2018?
- 2019?
- 2021?
- 2022?
- post-2023 rule environment?
- Power programs?
- Group-of-Five?
- favorites?
- underdogs?
- early lines?
- closing lines?
- different sportsbooks?

And most importantly:

Did we discover this pattern on the same observations used to test it?

If yes, it is not evidence yet.

Beware the conference-trend trap

If we test:

- 10 conferences,
- 12 months/weeks,
- home/road,
- favorite/underdog,
- high/low totals,
- weather combinations,
- kickoff times,
- rivalry status,

hundreds or thousands of “profitable systems” will appear by chance.

This is one of the biggest threats to the entire project.

Beat Vegas should distinguish:

Hypothesis discovered in training data

from

Hypothesis confirmed on untouched data.

Those should never be confused.

Comparing the three possible futures for Beat Vegas

Here is how I would assess the three options based on the current evidence.

Dimension	First-half Unders only	First-half Overs + Unders	Full-game Overs + Unders
Existing Beat Vegas knowledge	Highest	High	Medium
Historical market-data accessibility	Lower	Lower	Highest
Academic market research	Very limited	Very limited	Strongest
Model complexity	Lowest	Moderate	Highest
Game-state complexity	Lowest	Low	High
Opportunity count	Lowest	Higher	Highest if combined with 1H
Risk of directional bias	Highest	Lower	Lower

Dimension	First-half Unders only	First-half Overs + Unders	Full-game Overs + Unders
Ability to use weather/travel research	Moderate	Moderate	High
Ability to exploit early market	Moderate	Moderate	High
Ability to compare with strong closing benchmark	Harder historically	Harder historically	Much easier
Evidence market is broadly efficient	Unknown for 1H	Unknown for 1H	Yes
Evidence of specific inefficiencies	Sparse	Sparse	Yes
Data/engineering burden	Lowest	Moderate	Highest
Recommendation	Keep as champion	Best immediate challenger	Best second challenger / long-term expansion

Keeping Beat Vegas first-half Under only

There is a very legitimate argument for this.

Specialization is valuable.

You presumably already have:

- a grading methodology;
- a hypothesis about Under conditions;
- experience interpreting first-half environments;
- existing performance history;
- workflows designed around the market.

Throwing that away merely to generate more picks would be a mistake.

The largest weakness is philosophical: **Under-only architecture prevents the system from expressing the opposite conclusion even when the evidence says the market total is too low.**

It can also create confirmation bias. Every variable begins being interpreted through the question:

Why will this game go Under?

rather than:

What is the correct scoring distribution?

That distinction matters.

Verdict: Keep it, but stop treating “Under” as an immutable identity.

Expanding to first-half Overs and Unders

This is my preferred **first expansion**.

It forces Beat Vegas to become a pricing model rather than a filter model.

The engine would first answer:

Fair 1H total = 26.8

Then compare:

Market 30.5 → potential Under.

Market 24.5 → potential Over.

Market 27 → pass.

That “pass” is important.

A good system should conclude **no bet** on most games.

The advantage is that the infrastructure remains close to what you already understand. We can test whether the features driving your current Under selections actually predict first-half scoring errors—or merely happen to correlate with historical Under results.

It also gives us an elegant diagnostic:

If the model can successfully identify Unders but systematically fails on Overs, that tells us the existing Beat Vegas edge is **asymmetric**.

That would be an extremely valuable discovery.

Verdict: Highest-priority research expansion.

Moving into full-game Overs and Unders

This has the greatest long-term opportunity and the strongest research foundation.

Full-game lines have much better historical availability. Academic research has identified multiple conditional inefficiencies. Weather, travel, market movement, rule changes and team-total relationships all become highly relevant. ²

But this is also the market where we have strong evidence that the consensus becomes impressively accurate by closing time. ¹

So we should not confuse:

More games

with:

More edge.

A system placing 400 bets at 51.5% is worse than one placing 90 bets at a genuine 56%.

The goal is not action.

The goal is **expected value**.

Full games also introduce second-half game-state complexity, making them meaningfully different from your existing first-half problem.

Verdict: Absolutely worth building as a research challenger, but not worth replacing the existing system until it wins a head-to-head prospective test.

What Beat Vegas should ultimately become

The research changes the question.

I would no longer ask:

Should Beat Vegas become a full-game totals system?

I would ask:

Can Beat Vegas become a college-football scoring-market engine that knows which totals market is mispriced?

I think the answer is **yes, and that is the most promising direction**.

Conceptually, I would structure the future system like this:

Football layer

Team strength → matchup → expected possessions → scoring efficiency → weather → travel → personnel → coaching/scheme → rule environment.

↓

Scoring distribution

Expected Team A points
Expected Team B points
Expected 1H points
Expected full-game points
Uncertainty around each.

↓

Market layer

Full-game spread
Full-game total
Team totals
1H spread
1H total
Opening/current/closing prices
Cross-book prices.

↓

Mispricing layer

What is the sportsbook implying?

Where does Bet Vegas disagree?

How confident is that disagreement?

↓

Strategy layer

1H Under
1H Over
Full-game Under
Full-game Over
Eventually team totals.

↓

Decision

BET / PASS

That is fundamentally different from today's concept of "find first-half Unders."

The most important shift is from classification to pricing

I would move away from thinking:

Is this an Under game?

toward:

What should the total actually be, and how uncertain are we?

Example:

Beat Vegas expected 1H total: **27.1**
Estimated standard uncertainty: high
Sportsbook: **28**

Result:

PASS

Another:

Beat Vegas expected 1H total: **24.8**
Uncertainty: low
Sportsbook: **29.5 -110**

Result:

Strong Under candidate

Another:

Beat Vegas expected 1H total: **31.2**
Market: **27.5 -108**

Result:

Over candidate

That is how we remove directional bias.

I would make your existing first-half Under model the champion

Do not destroy something that may contain a real edge before proving the replacement.

Treat the existing strategy as:

Champion A — Beat Vegas 1H Under

Then research:

Challenger B — Beat Vegas 1H two-sided

and:

Challenger C — Beat Vegas full-game two-sided

The challengers do not win because their backtests look prettier.

They win only after performing better under the same standards.

I would require a challenger to demonstrate some combination of:

1. statistically credible out-of-sample advantage;
2. positive vig-free CLV;
3. profitable ROI after realistic prices;
4. stability across seasons;
5. stability after the 2023 timing regime;
6. sensible probability calibration;
7. performance that is not dependent on one tiny subgroup;
8. results that survive removal of suspicious variables.

Do not optimize for win rate

This may be the single most important philosophical change.

A 60% historical win rate can be meaningless if the model cherry-picked observations.

A genuine 54% process that:

- repeatedly beats sharp closes,
- survives multiple seasons,
- remains calibrated,
- works prospectively,
- and has hundreds of independent-ish observations

could be tremendously valuable.

The research on opening versus closing CFB lines makes this particularly important because the market itself provides us with an external benchmark against which Beat Vegas can be judged. ²⁰

The system should attempt to beat the market, not merely the scoreboard

Suppose we bet:

Under 58.5.

The game ends 72–21.

Terrible outcome.

But suppose the market closes:

54.5.

Our bet may have been excellent even though it lost.

Conversely:

We bet Under 55.5.

Market closes 61.5.

Final score is 17–10.

The bet won, but our process may have been wrong.

That is why Beat Vegas needs both:

Outcome grading

and

market grading.

Unabated's treatment of CLV makes the additional point that the comparison should ultimately be against a vig-free fair closing price, not merely whether our -110 wager looked better than another -110 number at close. ²¹

The strongest evidence-backed research targets are now clear

I would rank the Beat Vegas research agenda like this:

Highest priority

Weather × team style
Expected possessions/pace
Market opener versus current versus close
1H versus full-game implied scoring
Opponent-adjusted offensive/defensive efficiency
2023+ rule regime
Price-aware expected value.

Weather has direct CFB totals evidence; opening-to-close improvement has direct CFB evidence; timing rules have previously generated temporary market errors; and modern play-by-play data make possession-based modeling feasible. ²³

Second priority

Travel
Personnel/QB changes
Coordinator/scheme changes
Mismatch/game-state effects
FBS-FCS classification
starter removal/blowout behavior
team-total versus game-total inconsistencies.

Travel and team-total inconsistencies already have published empirical support; the others are highly plausible but need Beat Vegas-specific validation. ²⁴

Lower priority as standalone systems

Public betting percentages
Reverse line movement
conference Over/Under trends
rivalry trends
television status
recent Over/Under records.

Some of those contain useful information, but the available research gives little reason to believe they should serve as primary signals. In particular, RLM was not generally profitable in published college-football totals research, while sportsbooks appear capable of pricing around obvious Over preferences. ²⁵

My final verdict

There **can be a real edge in college-football totals**, but I would strongly reject the proposition that we have established a generic “college football totals are beatable” edge.

What we have established is more nuanced and more promising:

The overall market is strong. Full-game opening and closing totals have historically behaved approximately efficiently, and closing prices improve materially as bettors and information enter the market. ¹

The market is not omniscient. Peer-reviewed work has documented conditional problems involving weather, travel, rule changes and team-total structure. ²

The Over/Under direction itself is probably not the edge. Long-sample totals data are essentially 50/50 Under versus Over, and bettor preference for Overs appears known to sportsbooks. ²⁶

Early pricing is probably where information advantage is most monetizable. Professionals help move opening numbers toward a more accurate consensus, while opening limits are lower and closing lines historically outperform opening lines. ⁵

First-half totals remain a major unanswered question. I found insufficient published evidence to claim that your current market is objectively softer or sharper than the full-game market. That means your own historical and prospective Beat Vegas database can potentially answer a question the academic literature has not answered well.

And therefore:

I would not transition Beat Vegas from first-half Unders to full-game totals. I would transition Beat Vegas from an "Under system" into a "fair-price system."

The current first-half Under strategy remains the champion.

Build the research framework conceptually around a common scoring engine.

Allow it to estimate both Overs and Unders.

Test first-half both directions first.

Then test full-game totals independently.

Eventually compare game totals, first-half totals and team totals simultaneously.

Promote a market only when it demonstrates a real edge **out of sample and prospectively**, preferably while also beating a vig-free sharp closing benchmark.

The biggest potential competitive advantage is not predicting the most games correctly.

It is knowing:

which games the sportsbook has priced incorrectly, which portion of the game is mispriced, in which direction it is mispriced, by how much it is mispriced, when the

mispricing is largest, and whether that difference is large enough to bet after vig and uncertainty.

That is the version of Beat Vegas I believe has the best chance of becoming a genuinely defensible college-football betting system.

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